University of Toronto
Faculty of Arts and Science

MAT137Y Test 2

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Duration: 110 minutes
No Aids Permitted

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Please make sure your information above are correct.

This exam contains 10 **one-sided pages** (including this cover page) and is printed on 10 sheets of paper. There are 7 problems.

Instructions

- **Keep this booklet closed**, until an invigilator announces that the test has begun. However, you may fill out your information in this page before the test starts.
- Please place your **student ID** in a location on your desk that is easy for an invigilator to check without disturbing you during the test.
- **Turn your cell phone completely off** (not just in vibrate mode) and store it with your other belongs.
- **No aids are permitted on this examination.** Examples of illegal aids include but are not limited to textbooks, notes, calculators, cellphones, or any electronic device.
- If you need **space for your draft work**, you can use the back of each page as your scratch papers. Note that we will not scan the back of each page.
- **Write clearly** and concisely by using **dark pencil or pen.** Work scattered all over the page without a clear ordering will receive very little credit.
- For questions with a boxed area, ensure your answer is completely **inside the box**.
- **If you need more space**, use the blank pages at the end of the exam and clearly indicate on the question page when you have done this. **Do not tear any pages off this exam.**

Question:	1	2	3	4	5	6	7	Total
Points:	4	8	6	7	7	5	7	44
Score:								

Do not tear this page off. This page is a formula sheet and can be used for rough work only. It will not be graded under any circumstances.

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \sec\theta = \frac{1}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{1}{\tan\theta} \qquad \csc\theta = \frac{1}{\sin\theta}$$

$$\sin^2\theta + \cos^2\theta = 1 \qquad 1 + \tan^2\theta = \sec^2\theta \qquad 1 + \cot^2\theta = \csc^2\theta$$

$$\begin{aligned}\cos 2\theta &= \cos^2\theta - \sin^2\theta \\ &= 2\cos^2\theta - 1 \\ &= 1 - 2\sin^2\theta\end{aligned} \qquad \sin 2\theta = 2\sin\theta\cos\theta \qquad \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$$\cos^2\theta = \frac{1+\cos 2\theta}{2} \qquad \sin^2\theta = \frac{1-\cos 2\theta}{2} \qquad \tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$$

$$\sin A\sin B = \frac{1}{2}\big(\cos(A-B) - \cos(A+B)\big) \qquad \sin(A+B) = \sin A\cos B + \cos A\sin B$$

$$\cos A\cos B = \frac{1}{2}\big(\cos(A-B) + \cos(A+B)\big) \qquad \cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\sin A\cos B = \frac{1}{2}\big(\sin(A+B) + \sin(A-B)\big) \qquad \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

$$c^2 = a^2 + b^2 - 2ab\cos C \qquad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

You may use the rest of this formula sheet page for rough work. It will not be graded. If you need more space for a solution, please use one of the blank pages at the back of the test.

1. For each of the following questions, only your final answer will be graded.

No justification is necessary.

(1a) (1 point) Compute $\lim_{x\to-\infty} \dfrac{x^3+\sqrt{4x^6+2}}{x^2+3x^3-2x}$.

Final Answer

$-\frac{1}{3}$

(1b) (1 point) Let $f(x) = x^2\ln(e^{2x}+1)$ and calculate $f'(x)$.

Final Answer

$2x\ln(e^{2x}+1) + x^2\frac{2e^{2x}}{e^{2x}+1}$

(1c) (1 point) Let $f(x) = \arctan(\sin(x)+3x+1)$ and calculate $f'(0)$.

Final Answer

2

(1d) (1 point) Estimate $e^{0.01}$ by using a linear approximation.

Final Answer

1.01

For each of the following questions, only your final answer will be graded.

No justification is necessary.

2. (2a) (2 points) Let $f(x)$ be an one-to-one function on $\mathbb{R}$ and define its inverse to be f^{-1}. We know the tangent line of $f(x)$ at 2 is $y = 3x - 2$. Let $g(x) = (f^{-1}(x) + 1)^2$. Find $g(4)$ and $g'(4)$.

Final Answer

$g(4) =$ 9

Final Answer

$g'(4) =$ 2

(2b) (2 points) Let $f(x) = (\cos^2(x) + 1)^x$. Compute $f'(x)$.

Final Answer

$(\cos^2(x)+1)^x\left[(\ln(\cos^2(x)+1)) - 2x\tan x\right]$

(2c) (2 points) A curve satisfies the equation $2x^2y = y^3 + x^2 + 3$. Define a function $y = f(x)$ near $(1, -2)$. Find the equation of the tangent line of $f(x)$ at $(1, -2)$.

Final Answer

$-x-1$

(2d) (2 points) State the Intermediate Value Theorem. Make sure to specify all assumptions.

Final Answer

Assuming $a, b \in \mathbb{R}$, $\exists M \in \mathbb{R}$

If a function f is defined and continuous on $[a,b]$ and if $f(a) < M < f(b)$,

$\exists c \in \mathbb{R}$, st $f(c) = M$

For each of the following questions, only your final answer will be graded.

No justification is necessary.

3. For each of the statement below, construct one function f with domain $\mathbb{R}$ such that f satisfies the given statement.

(3a) (2 points) f is (strictly) increasing on $\mathbb{R}$ and $f'(2) = 0$.

Final Answer

$f(x) = (x-2)^3$

(3b) (2 points) f is continuous on $\mathbb{R}$, f has a local max at $x = 0$, and $f'(0)$ does not exist.

Final Answer

$f(x) = -|x|$

(3c) (2 points) f is differentiable at 0 but $f \circ f$ is not differentiable at 0.

Final Answer

$f(x) = |x-1|$

4. Let $f(x) = \sin x$.

(4a) (1 point) Find the largest interval containing 3 such that f is invertible on this interval.

Final Answer

$[\pi/2, 3\pi/2]$

(4b) (1 point) Define h to be the inverse function of f restricted on this interval. Which is the domain of this inverse function h.

Final Answer

$[-1, 1]$

(4c) (1 point) Find $f(h(0.6))$.

Final Answer

0.6

(4d) (1 point) Find $h(f(6))$.

Final Answer

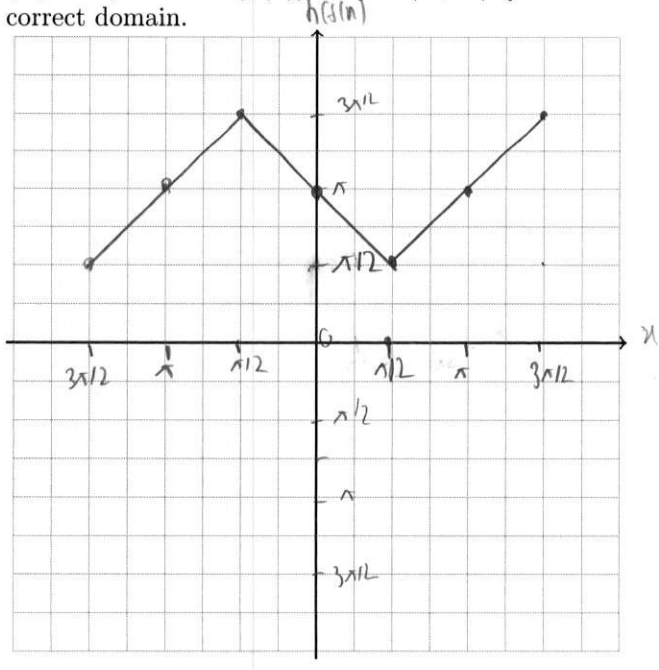
$3\pi - 6$

(4e) (1 point) Find h in terms of arcsin.

Final Answer

$\arcsin 6 + \pi$

(4f) (2 points) Sketch $h(f(x))$ on $[-3\pi/2, 3\pi/2]$. Make sure to label your axes and that you have the correct domain.



5. (5a) (1 point) Let $a \in \mathbb{R}$. Write the limit definition of "f is differentiable at a".

For a function f defined on an interval centred a, we say "f is differentiable on a", when $\lim_{x \to a} = \frac{f(x)-f(a)}{x-a}$

(5b) (6 points) Assume that $a > 0$. Let f be a function defined on $\mathbb{R}$ that is differentiable at a. Define g by $g(x) = \dfrac{f(x)}{\sqrt{x}}$ for all $x > 0$. Prove that g is differentiable at a and that

$$g'(a) = \frac{\sqrt{a}f'(a) - \frac{f(a)}{2\sqrt{a}}}{a}.$$

Write a proof directly from the definition of the derivative as a limit without using any of the other differentiation rules, e.g. quotient rule or chain rule. No epsilon-delta proof is needed for this question. *Hint: you may need to use $x - a = (\sqrt{x} - \sqrt{a})(\sqrt{x} + \sqrt{a})$ in your proof.*

- Assume $a > 0$
- Since differentiability implies continuity, $f(x)$ is continuous at a as well
- $\sqrt{x}$ is a continuous function and it is differentiable, so $\lim_{x \to a} f(x) = fa$
- Since $f(x)$ and $\sqrt{x}$ are both differentiable at continuous, $g(x)$ is also differentiable and continuous

Proof:

$$g'(a) = \lim_{x \to a} \frac{g(x) - g(a)}{x-a}$$

$$= \lim_{x \to a} \frac{\frac{f(x)}{\sqrt{x}} - \frac{f(a)}{\sqrt{a}}}{x-a}$$

$$= \lim_{x \to a} \left(\frac{\sqrt{a}f(x) - f(a)\sqrt{x}}{\sqrt{x}\sqrt{a}\,(x-a)} \right)$$

$$= \lim_{x \to a} \left(\frac{\sqrt{a}f(x) - \sqrt{a}f(a) + \sqrt{a}f(a) - f(a)\sqrt{x}}{\sqrt{x}\sqrt{a}\,(x-a)} \right)$$

$$= \lim_{x \to a} \left(\frac{\sqrt{a}(f(x) - f(a)) - f(a)(\sqrt{x} - \sqrt{a})}{\sqrt{x}\sqrt{a}\,(x-a)} \right)$$

$$= \lim_{x \to a} \left(\frac{\sqrt{a}}{\sqrt{x}\sqrt{a}} \cdot \frac{(f(x)-f(a))}{x-a} - \frac{f(a)(\sqrt{x}-\sqrt{a})}{\sqrt{x}\sqrt{a}(x-a)} \right)$$

Using limit laws and $x-a = (\sqrt{x}-\sqrt{a})(\sqrt{x}+\sqrt{a})$, we gets

$$g'(a) = \lim_{x \to a} \left(\frac{\sqrt{a}}{\sqrt{x}\sqrt{a}} \right) \cdot \lim_{x \to a} \frac{(f(x)-f(a))}{x-a} - \lim_{x \to a} \frac{f(a)(\sqrt{x}-\sqrt{a})}{\sqrt{x}\sqrt{a}(\sqrt{x}-\sqrt{a})(\sqrt{x}+\sqrt{a})}$$

$$= \frac{\sqrt{a}}{a} \cdot f'(a) - \lim_{x \to a} \frac{f(a)}{\sqrt{x}\sqrt{a}(\sqrt{x}+\sqrt{a})}$$

$$= \frac{\sqrt{a}}{a} f'(a) - \frac{f(a)}{\sqrt{a}\sqrt{a}(\sqrt{a}+\sqrt{a})}$$

$$= \frac{\sqrt{a}f'(a)}{a} - \frac{f(a)}{2a\sqrt{a}}$$

$$= \frac{1}{a}\left(\sqrt{a}f'(a) - \frac{f(a)}{2\sqrt{a}}\right)$$

$$= \left(\sqrt{a}f'(a) - \frac{f(a)}{2\sqrt{a}}\right) \div a$$

$$= \frac{\sqrt{a}f'(a) - \frac{f(a)}{2\sqrt{a}}}{a}$$

6. (5 points) Let $f(x) = \dfrac{1}{3+x^4}$. We say that a rectangle R is inscribed between the graph of f and the x-axis, if two top vertices of R are on the graph of f and two bottom vertices are on the x-axis. What is the largest possible area of a rectangle inscribed between the graph of f and the x-axis? Show all of your work including justifying your answer is the largest possible area.

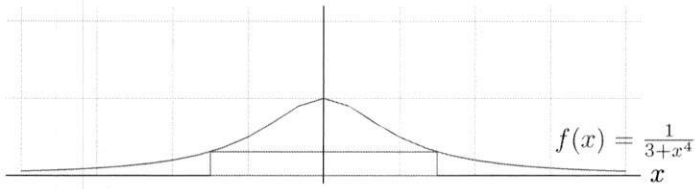


Figure 1: An example of an inscribed rectangle in red

Let x_1 be x coordinates on right side of y axis

Let x_2 be x coordinates on left side of y axis

Let $|x_1| = |x_2| = x$

Length is $|x_1| + |x_2| = 2x$

Area = length × width $= 2x \times \frac{1}{3+x^4}$

So $A = \dfrac{2x}{3+x^4}$

$$\frac{dA}{dx} = \frac{(3+x^4)\times 2 - 4x^3(2x)}{(3+x^4)^2}$$ (quotient rule)

$$= \frac{6+2x^4-8x^4}{(3+x^4)^2}$$

$$= \frac{-6(x^4-1)}{(3+x^4)^2}$$

For a maximum/minimum areas $\frac{dA}{dx} = 0$

So $\dfrac{-6(x^4-1)}{(3+x^4)^2} = 0$

$-6(x^4-1) = 0$

$-6(x^2-1)(x^2-1) = 0$

$-6(x^2+1)(x+1)(x-1) = 0$

$-(x^2+1)$ has no solutions

So $(x+1) = 0$, $(x-1) = 0$

$x_2 = -1$, $x_1 = 1$

Area $= \dfrac{2(1)}{3+1^4} = \dfrac{2}{4} = \dfrac{1}{2}$ units^2

7. (7a) (2 points) State the Mean Value Theorem.

Assuming $a, b \in \mathbb{R}$ and with $a < b$, if there is a function 'f' that is:

1) Differentiable on (a, b)
2) Continuous on $[a, b]$

Then:

$$\exists c \in (a, b) \text{ st } f'(c) = \frac{f(b) - f(a)}{b - a}$$

(7b) (5 points) Consider the function $f(x) = \ln(x)$ on the interval $[1/3, 3]$. Prove that

$$\forall x_1, x_2 \in [1/3, 3],\ |f(x_1) - f(x_2)| \leq 3|x_1 - x_2|$$

According to EVT, since $f(x)$ is continuous, the function has a maximum and minimum within $[1/3, 3]$

$f'(\frac{1}{3}) = \frac{1}{x} = \frac{1}{\frac{1}{3}} = 3 \quad f'(3) = \frac{1}{3}$

Within this interval $f'(c) \leq 3$

~~Sa~~ According to mean value theorem, if f is differentiable on (a, b), continuous on $[a, b]$, $\forall x_1, x_2 \in \mathbb{R}$, $[x_1, x_2] \in [a, b]$, if function defined on $[x_1, x_2]$

$$f'(c) = \frac{f(x_1) - f(x_2)}{x_1 - x_2}$$

Since function is always positive, we can say

$$f'(c) = \frac{|f(x_1) - f(x_2)|}{|x_1 - x_2|} \leq 3$$

$$|f(x_1) - f(x_2)| \leq 3|x_1 - x_2|$$

Do not tear this page off. This page is for additional work and will not be marked, unless you clearly indicate it on the original question page.